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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B.Tech (CBCS) Degree Examinations

December 2024

EE2201-1 – POWER SEMICONDUCTOR DEVICES

(For EE)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

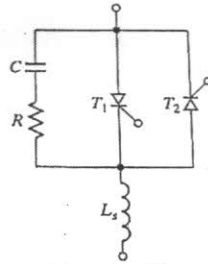
Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. Which among the following is an example for a voltage bidirectional 2 quadrant switch ?
A) SCR
B) BJT
C) Diode
D) MOSFET
2. The unit of the apparent power is
A) Watts
B) VAR
C) VA
D) None
3. How diode circuit behaves in reverse mode ideally?
A) Resistive
B) Short circuit
C) Open circuit
D) Series
4. In an ideal diode, the current flows in _____ ?
A) Unidirectional
B) Bidirectional
C) Multiple directions
D) Reverse direction
5. In a power MOSFET, the term “threshold voltage” (V_{th}) refers to
A) the maximum gate- source voltage before breakdown
B) the voltage at which channel is formed
C) the minimum voltage required to fully turn off MOSFET
D) the maximum drain- source voltage rating
6. At turn-on the initial delay or turn on delay is the time required for the
A) input inductance to charge to the threshold value
B) input capacitance to discharge to the threshold value
C) input inductance to discharge to the threshold value
D) input capacitance to charge to the threshold value
7. Which among the following devices is the most suited for high frequency applications?
A) BJT
B) IGBT
C) MOSFET
D) SCR
8. The controlling parameter in MOSFET is
A) V_{ds}
B) I_g
C) I_D
D) V_{gs}
9. What is the primary function of the gate in an IGBT?
A) To control the current flow between the collector and the emitter
B) To act as a power dissipation element
C) To provide a path for the base current
D) To isolate the collector and emitter terminals

10. **What is the primary advantage of an IGBT over a power MOSFET?**
A) Higher switching frequency B) Lower on-state voltage drop
C) Lower gate drive voltage D) Higher input capacitance
11. **What typically triggers latch-up in an IGBT?**
A) Excessive gate voltage B) High collector current
C) High temperature conditions D) Overvoltage or high-current pulses
12. **When designing an over-current protection scheme, why is it important to consider the IGBT's safe operating area (SOA)?**
A) To ensure the IGBT operates at higher efficiency B) To ensure the IGBT does not exceed its maximum current and voltage ratings
C) To improve the switching frequency of the IGBT D) To enhance thermal management
13. **In which type of application are inverter-grade thyristors most commonly used?**
A) Low-frequency AC motor drives B) High-frequency inverters and power converters
C) Analog signal processing D) Digital signal switching
14. **What distinguishes inverter-grade thyristors from standard thyristors?**
A) Higher gate current B) Better thermal management
C) Higher voltage and current ratings, and enhanced switching performance D) Lower turn-off time
15. **Which of the following wide bandgap semiconductors is commonly used for high-frequency applications?**
A) Silicon Carbide (SiC) B) Germanium (Ge)
C) Silicon (Si) D) Gallium Nitride (GaN)
16. **Which of the following is a key challenge associated with wide bandgap semiconductors?**
A) Low breakdown voltage B) High cost and complex manufacturing processes
C) Low thermal stability D) Limited frequency response
17. **Which of the following is a common application for electrical isolation in power electronics?**
A) To amplify analog signals B) To separate high voltage sections from low voltage control circuits
C) To improve the frequency response of the circuit D) To increase the thermal conductivity of components
18. **What is the primary function of a pulse transformer in isolation?**
A) To filter high-frequency noise B) To amplify weak signals
C) To regulate the voltage level D) To provide galvanic isolation while transferring digital signals
19. **Why is a specialized gate drive circuit often required for SiC and GaN devices?**
A) To reduce the size of the semiconductor B) To handle high switching speeds and high gate charge
C) To lower the breakdown voltage of the devices D) To increase the thermal resistance
20. **What is the primary purpose of a snubber circuit in power electronics?**
A) To amplify the input signal B) To filter high-frequency noise
C) To protect switching devices from voltage spikes and reduce ringing D) To increase the power factor of the circuit

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Give the classification of switches with respect to controllability.	4	L*1	1	1,2
	b) Describe the power loss details in switches.	4	L2	1	1,2
	c) Explain the basic structure & operation of power MOSFET.	8	L2	2	1,2
2.	a) List the merits of power electronics as compared to conventional methods of power processing.	4	L1	1	1,2
	b) Justify the use of idealized device characteristics in analyzing converter topologies.	4	L3	1	1,2
	c) Explain the MOSFET switching model with a neat waveform.	8	L2	2	1,2
3.	a) Explain the i-v & idealized characteristics of Diode, also list the types of diode.	8	L2	1	1,2
	b) Explain with a sample circuit, how a power MOSFET is protected against high di/dt and dv/dt.	8	L2	2	1,2
Unit – II					
4.	a) With a neat figure explain the structure of IGBT with equivalent circuit diagram.	8	L2	3	1,2
	b) Explain the construction & features of phase controlled thyristor.	8	L2	4	1
5.	a) Describe turn on & turn off switching characteristics of IGBT.	8	L2	3	1,2
	b) Explain the constructional feature of inverter graded thyristor.	8	L2	4	1
6.	a) Explain the causes of latch up in IGBT? How it can be avoided.	8	L2	3	1,2
	b) Give comparison between MOSFET & IGBT.	8	L2	4	1
Unit – III					
7.	a) Explain the necessity of isolation of gate & base drive circuits, explain any one method.	8	L2	5	1,2,3
	b) An RC snubber circuit, as shown in fig.7b has $C = 0.75\mu\text{F}$, $R = 6.6\Omega$ & input voltage $V_s = 220\text{V}$. The circuit inductance is $L_s = 50\mu\text{H}$. Determine i) the peak forward voltage V_p , ii) the initial dv/dt iii) the maximum dv/dt.				
 Figure. 7b		8	L3	5	1,2,3
8.	a) What are the precautions to be taken in mounting a device on a heat sink?	8	L2	5	1,2,3
	b) Design a snubber circuit for the following data:				
	i) Supply transformer is rated 5kVA with secondary voltage of 120V RMS				
	ii) Switching frequency = 400Hz				
	iii) Circuit inductance = 100μH				
	Peak transient voltage is to be limited at 200volts, (v) $\delta = 0.75$	8	L4	5	1,2,3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
